

CO4 SIMULATION TASK

PROGRAM CODE :

```
clc; clear; close all;

% ----- % RECTANGULAR WAVEGUIDE - X
BAND % TE10 MODE CUTOFF FREQUENCY % -----
-

% Constants c = 3e8; % Speed of light (m/s)

% Waveguide dimensions a = 22.86e-3; % Width (m) b = 10.16e-3; % Height (m)

% TE10 mode m = 1; n = 0;

% Cutoff frequency calculation  $f_c = (c/2) * \sqrt{(m/a)^2 + (n/b)^2}$ ;

% X-band frequency range f = linspace(8e9,12e9,1000);

% Normalized frequency  $f_n = f/1e9$ ;

% Calculate propagation constant  $\beta = (2\pi f/c) .* \sqrt{1 - (f_c/f).^2}$ ;

% Display results fprintf('=====\n');
fprintf(' RECTANGULAR WAVEGUIDE - X BAND\n');
fprintf('=====\n'); fprintf('Waveguide width
a = %.2f mm\n',a/1000); fprintf('Waveguide height b = %.2f mm\n',b/1000); fprintf('Mode =
TE10\n'); fprintf('Cutoff frequency = %.3f GHz\n',f_c/1e9); fprintf('X-band range = 8 - 12
GHz\n'); fprintf('=====\n');

if f_c < 8e9 fprintf('RESULT: TE10 mode propagates throughout X-band.\n'); else
fprintf('RESULT: TE10 mode does not propagate throughout X-band.\n'); end

% ----- % GRAPH 1: Propagation constant versus
frequency % -----

figure;

plot(f_n,beta,'LineWidth',2); hold on;
```

```

xline(fc/1e9,'--','LineWidth',2); xline(8,'--','LineWidth',1.5); xline(12,'--','LineWidth',1.5);

grid on;

xlabel('Frequency (GHz)'); ylabel('Propagation Constant \beta (rad/m)'); title('TE_{10} Mode
Propagation in X-Band');

legend('\beta for TE_{10}', ... 'TE_{10} Cutoff Frequency', ... 'X-Band Lower Limit', ... 'X-Band
Upper Limit', ... 'Location','northwest');

xlim([5 13]);

% ----- % GRAPH 2: Normalized propagation
factor % -----

figure;

prop_factor = sqrt(1-(fc./f).^2);

plot(fn,prop_factor,'LineWidth',2); hold on;

xline(fc/1e9,'--','LineWidth',2);

grid on;

xlabel('Frequency (GHz)'); ylabel('Normalized Propagation Factor'); title('TE_{10} Mode
Propagation Factor');

legend('sqrt(1-(f_c/f)^2)', ... 'Cutoff Frequency', ... 'Location','southeast');

xlim([6 13]); ylim([0 1]);

```

OUTPUT:

```
=====
RECTANGULAR WAVEGUIDE - X BAND
=====
Waveguide width a = 22.86 mm
Waveguide height b = 10.16 mm
Mode               = TE10
Cutoff frequency   = 6.562 GHz
X-band range       = 8 - 12 GHz
=====
RESULT: TE10 mode propagates throughout X-band.
>> |
```



